

Learner's AI

Using Machine Learning to empower Human Learning

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The Evolution of Digital Learning

Understanding Modern Learners and System Challenges

Current Toolkit: Learning Management Systems (LMS)

Core LMS Features



Examples: Moodle,
Canvas, Blackboard

-  Centralized course management
-  Administration and progress tracking
-  Resource access and distribution
-  Assignments and assessments
-  Discussion forums and collaboration tools
-  Integration with SIS & CRM systems

Bridging the Gap: Challenges in Digital Learning



Ineffective Assessments

Assessment methods may not accurately reflect true understanding or skills.



Lack of Personalization

One-size-fits-all approaches fail to cater to individual learner needs.



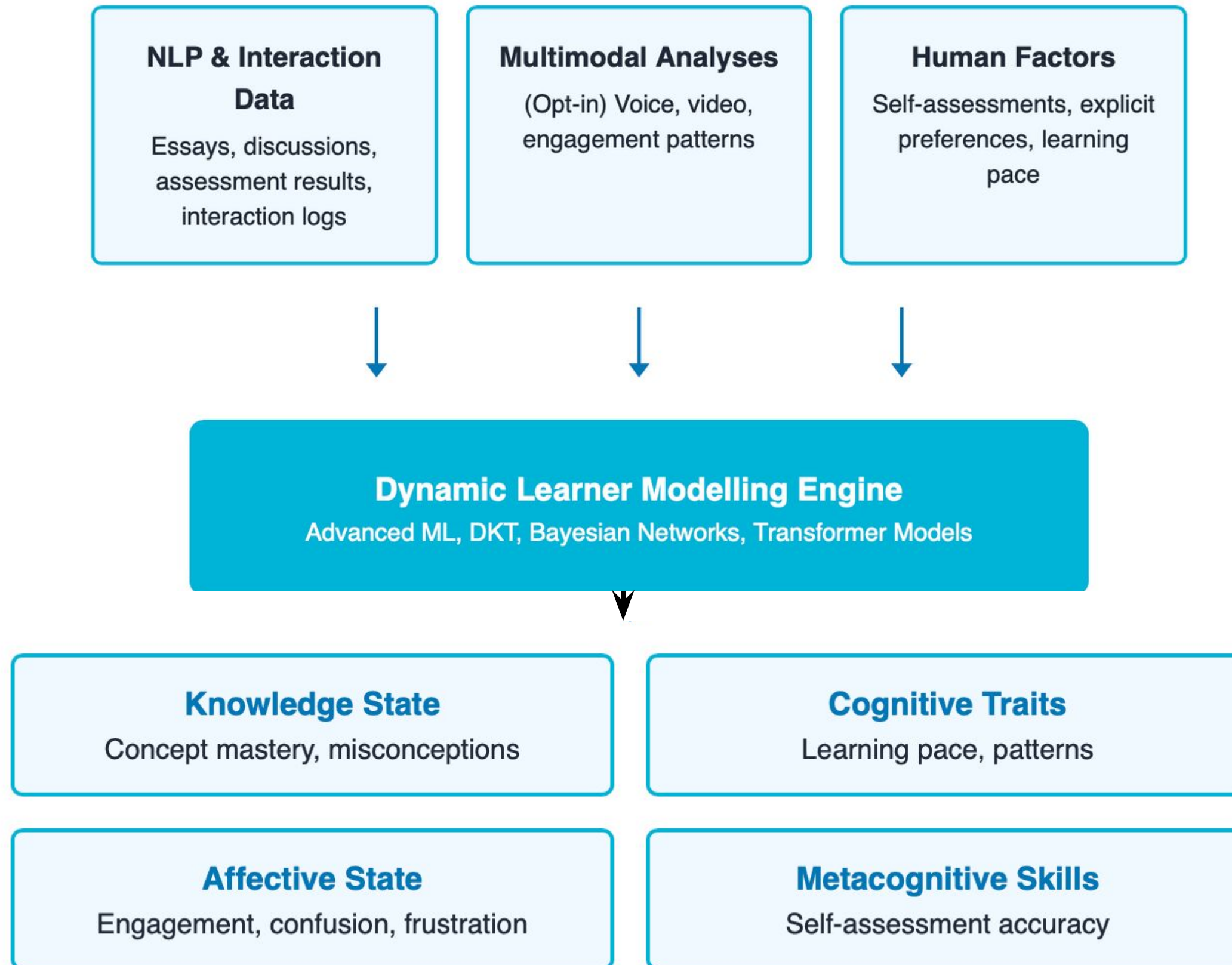
Poor Interaction & Feedback

Limited meaningful engagement and delayed or generic feedback.

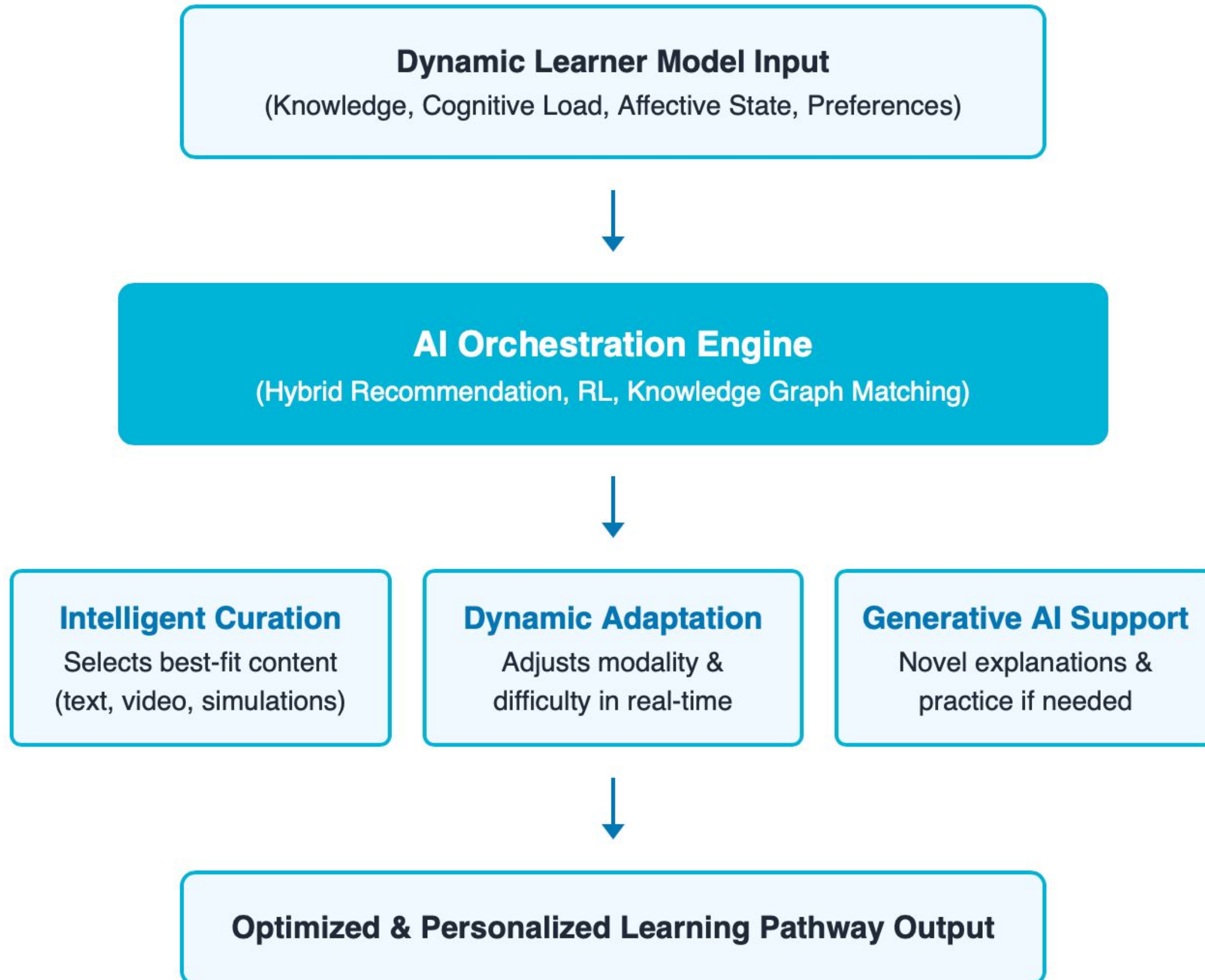
Technology Description

1. Dynamic Learner Modelling Engine

Engine Architecture: From Data to Insight



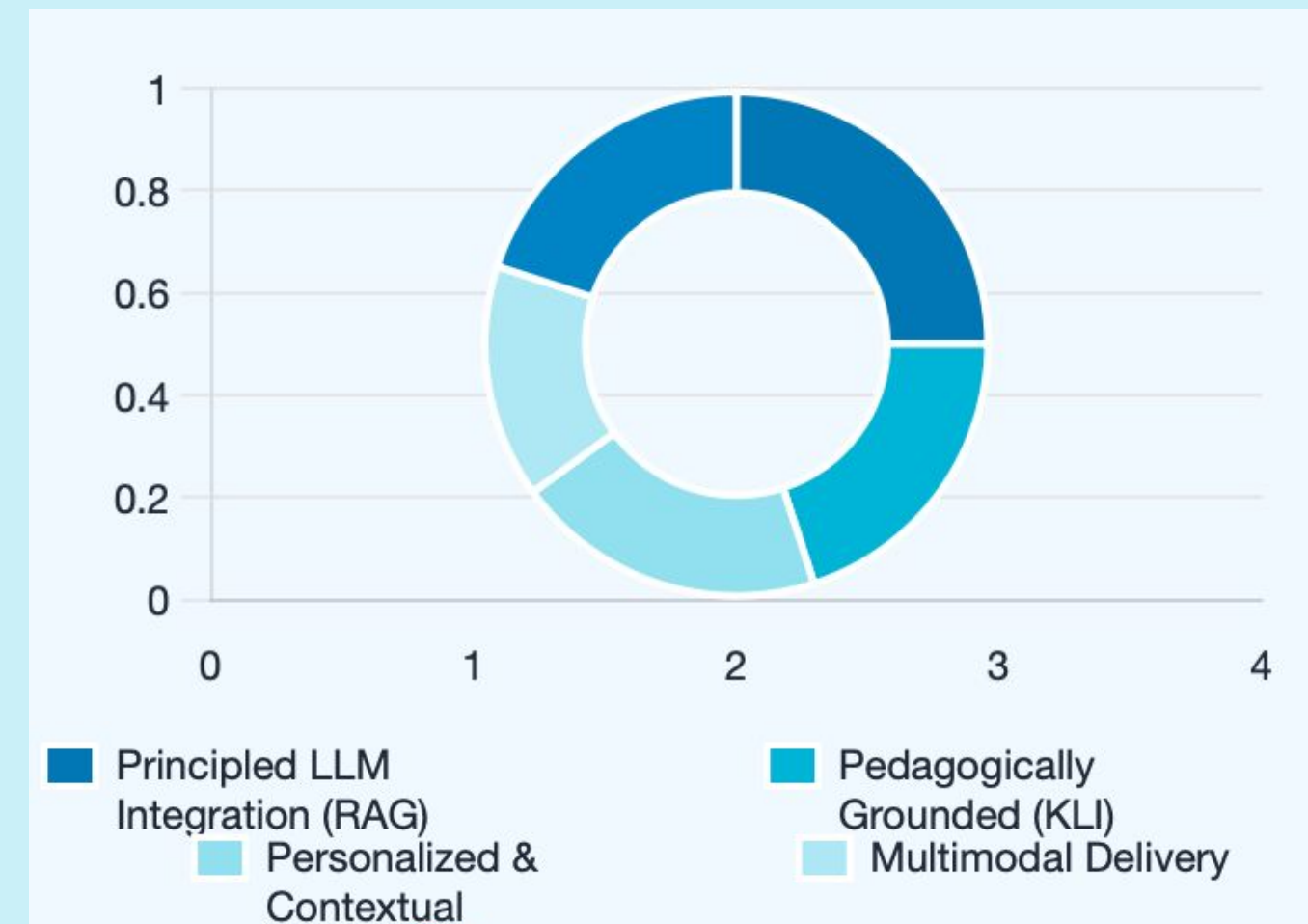
2. AI Orchestration Engine



3. Advanced Intelligent Feedback

Key Pillars of Our Feedback System:

- **Principled LLM Integration:** Carefully designed, contextualized, and evaluated LLM outputs, grounded by Retrieval Augmented Generation (RAG).
- **Theoretically Grounded Content:** Feedback guided by learning science (e.g., KLI framework) for memory, problem-solving, and deep understanding.
- **Personalized & Contextual:** Tailored to student's error, learner model, and emotional state.
- **Multimodal Delivery:** GenAI-powered feedback via text, images, audio, or short video explanations.
- **Scaffolding & Metacognition:** Interactive support that fades with proficiency, promoting self-correction and reflection.



Key Research Achievements & Validations

Showcasing validated advancements in AI-driven personalized learning.

Current Progress



Advanced Emotion Recognition

Subtitle: For Cognitive-Affective State Modelling from ITS Conversations.

Achievement: Computationally efficient AI for student emotion understanding (audio + text).

Technology: Bidirectional Transformers with attention-based multimodal fusion.

Validation:

~62.09% WF1

Comparable to SOTA on MELD dataset (Farooq et al., 2024). Crucial for adaptive systems.



NLP-Powered Content & Assessment

Subtitle: Automated pipelines for dynamic learning resource creation.

Key Pipelines: Learning Pathway Segmentation, Definition Generator (GPT-2), Overview & Reflection Quiz Generation.

Validation (Definition Generator):

High ROUGE 0.2565

Fine-tuned GPT-2 significantly outperforms baselines (Diwan et al., 2023). Optimal temperature (0.4) identified.



Bayesian Modelling of Human Factors

Subtitle: Optimizing learning experiences by understanding cognitive variables.

Objective: Identify optimum levels of cognitive variables for ideal learning.

Future: Extend model for more cognitive variables.

Initial Validation:

**"Combined" Model
Ranked Highest**

Successfully identified ideal student confidence/expectation levels for language tasks.

Future Roadmap



Months 1-3

Foundational AI & Core Modeling Enhancement

Focus: Solidify and expand current AI progress.

- V1 Dynamic Learner Model
- Advanced Content Processing
- Data Parameterization



Months 4-6

Developing Core ITS Features & Feedback Framework

Focus: Building initial versions of differentiating ITS features, particularly intelligent feedback and personalization logic.

- Feedback Framework Design (Principled LLM Integration)
- Personalized Pathway Logic (Alpha)
- Enhanced MCQ Generator Refinement

Future Roadmap



Months 7-8: PoC/MVP Launch

PoC/MVP Development, Integration & Launch

Focus: Integrating core components into a demonstrable PoC and a usable MVP.

Key Milestone: PoC/MVP Launch (End of Month 8)

- V1 Intelligent Feedback System (MVP Feature)
- Core Component Integration (MVP Feature)
- Basic Personalized Content Delivery (MVP Feature)
- Internal PoC/MVP Launch & User Testing Readiness



Months 9-12

Post-MVP Iteration, Enhancement & Pilot Preparation

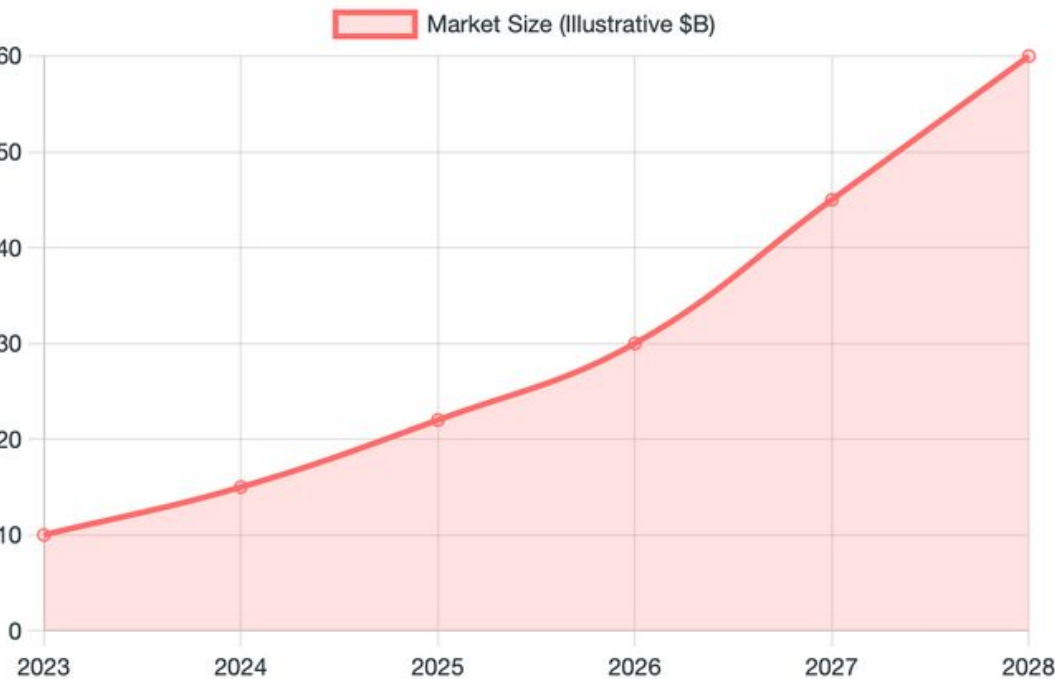
Focus: Iterating on the MVP based on internal feedback, adding depth to features, and preparing for initial external pilot programs.

- MVP Refinement (Stability, Usability)
- Expand Learner Modeling (Cognitive traits, preferences)
- Broaden Feedback Suite (Sophistication, multimodal)
- LMS Integration (Beta Design - Moodle)
- Pilot Program Preparation

Business Model

Projected Growth of AI in Education

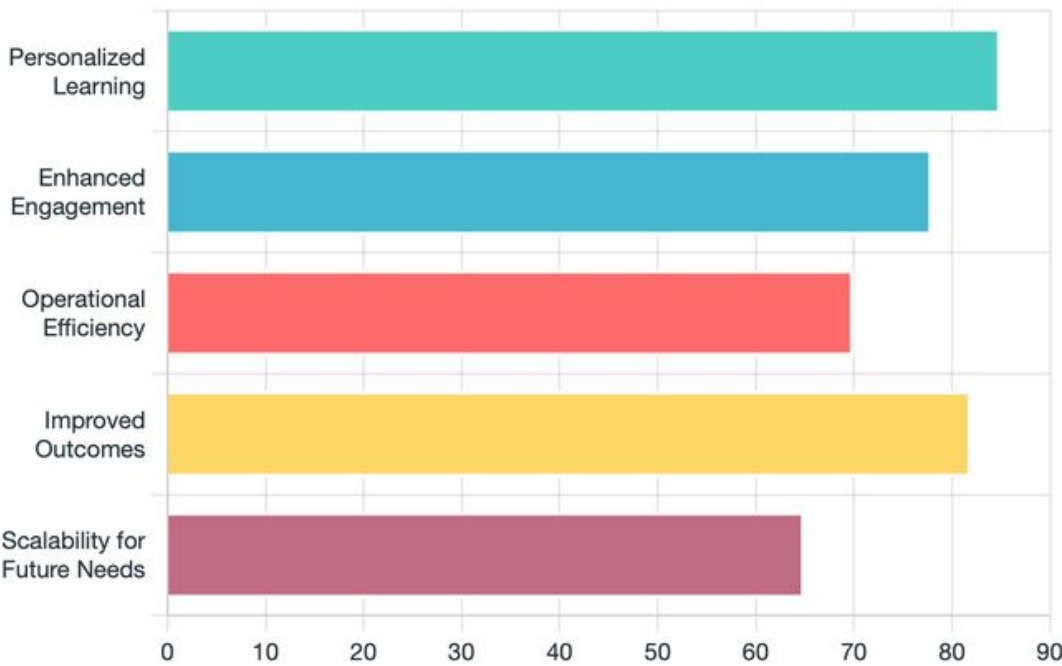
Illustrative trend showing the expanding market for AI-driven learning solutions.



\$USD 33.2 billion by 2033.

Key Drivers for AI Learning Adoption

Factors influencing institutions to adopt solutions like Learner's.ai.



💡 Value Proposition



Personalized Learning Paths: Tailored educational journeys adapting to individual student needs and pace.



Engaging Content: Interactive and captivating materials designed to maintain student interest and improve knowledge retention.



Intelligent Feedback: Smart, actionable insights for learners, helping them understand progress and areas for improvement.



Easy LMS Integration: Seamless connection with existing Learning Management Systems for hassle-free adoption.

🎯 Target Audience



Educational Institutions (K-12, Higher Ed)



Corporate Training Programs

We cater to diverse learning environments, providing tools that enhance traditional education and professional development.

💰 Revenue Streams & 📢 Channels

Revenue Streams

- **Subscription-Based Model:** Ensuring predictable, recurring revenue.
- **Tiered Pricing:** Flexible options (per student or institutional packages) to cater to different needs and budgets.

Channels

- **Direct Sales to Institutions:** Building strong relationships with educational decision-makers.
- **LMS Provider Partnerships:** Leveraging existing networks to expand reach and simplify integration.

Go-To-Market Strategy



Phase 1: VALIDATE

Laying the Foundation

Action:

- Launch pilot programs with select educational institutions.

Goal:

- Deeply embed solution and refine product based on real-world insights.
- Gather undeniable proof of concept (personalized paths, engagement, intelligent feedback).
- Build compelling case studies and identify early champions/advocates.



Phase 2: SCALE

Expanding Our Reach

Action:

- Deploy expert direct sales team armed with success stories.
- Forge strategic partnerships with LMS providers (leveraging easy integration).

Goal:

- Expand regionally, replicating proven success from pilot programs.
- Showcase clear ROI and superior learning experiences to institutions.
- Utilize flexible subscription & tiered pricing models for accessibility.



Phase 3: DOMINATE

Leading the Market

Action:

- Strategically expand into new high-growth markets (e.g., corporate training).
- Partner with major MOOC platforms (e.g., Coursera, Udemy).

Goal:

- Achieve market leadership through continuous improvement and broader reach.
- Relentlessly pursue product evolution to deliver cutting-edge AI and value.
- Ensure our AI stays at the forefront, constantly enhancing learner value.

Why the project deserves the Award



Innovation at its Core

- ✓ Dynamic Learner Modeling Engine (deep student understanding).
- ✓ Multimodal Emotion Recognition (ERC) system developed.
- ✓ Adaptive Content & Personalized Pathway Orchestration (NLP tested).
- ✓ Improved Automatic MCQ Generator.
- ✓ Advanced Intelligent Feedback & Scaffolding via Principled LLM Integration



Expected Impact

- ✓ Dramatically improve learning outcomes and reduce student frustration.
- ✓ Foster a genuine love for learning through personalized, adaptive, and engaging experiences.
- ✓ Make learning truly supportive and effective for a diverse range of students.
- ✓ Contribute significantly to the advancement of educational technology.



Entrepreneurial Vision

- ✓ Driven to address shortcomings in current digital learning.
- ✓ Student-centric, design-thinking approach to solve real-world problems.
- ✓ Ultimate Goal: Incubate technology & build a sustainable venture.
- ✓ Passionate about creating tangible educational value.



Investment Potential

- ✓ Addresses critical need in expanding Indian & global EdTech markets.
- ✓ Attractive proposition for future investment and growth.
- ✓ Strong Differentiated technology Competitive Edge: (comprehensive learner modeling, principled AI feedback).

Academic validation

Re: Extending AI Placebo Research: Bayesian Re-analysis of Kosch et al.'s (2022) study



From Thomas Kosch <thomas.kosch@hu-berlin.de> on 28.05.2025 00:36

[Details](#)

Dear Utkarsh,

Thank you for your message and for sharing your report and slides. I was really surprised and impressed to see that our earlier work inspired such a detailed and thoughtful re-analysis :). Your use of hierarchical Bayesian modeling to explore how

Thank You!